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**Sayed Rezwanul Islam Biplab,
Md. Hasan Ali, Md. Mahabub Alam
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Rahman & Jaker Hossain**

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Performance enhancement of CIGS-based solar cells by incorporating an ultrathin BaSi₂ BSF layer

Sayed Rezwanul Islam Biplab¹ · Md. Hasan Ali¹ · Md. Mahabub Alam Moon¹ · Md. Firoz Pervez² · Md. Ferdous Rahman^{1,3} · Jaker Hossain³

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Abstract

Conventional copper indium gallium diselenide (CIGS)-based solar cells offer higher efficiency than other second-generation technologies such as hydrogenated amorphous silicon (a-Si:H)- or cadmium telluride (CdTe)-based solar cells, but higher manufacturing cost due to the use of the rare metals indium and gallium. The purpose of the work presented herein is to improve the efficiency of such devices by using cheaper materials. Accordingly, a back-surface field layer made of low-cost and widely available barium silicide (BaSi₂) with a thickness of 0.3 μm is introduced for the first time into the basic CIGS solar cell structure consisting of Al/ZnO/CdS/CIGS/Mo, resulting in the alternative structure of Al/FTO/CdS/CIGS/BaSi₂/Mo, with fluorine-doped tin oxide (FTO) as the window layer. One-dimensional simulations of the solar cell capacitance are employed to study the photovoltaic parameters such as the power conversion efficiency, short-circuit current density, open-circuit voltage, fill factor, and quantum efficiency of the devices. The thickness of the CIGS absorber layer is varied from 0.1 to 3 μm to optimize the device. Besides, the effects of the acceptor ion and bulk defect densities in the CIGS absorber layer, cell resistances, and operating temperature on the overall performance are also investigated. The proposed structure offers an efficiency of 26.24% with a thin CIGS layer of only 0.8 μm . In addition to reduced CIGS thickness and cost, the presented approach results in CIGS solar cells with enhanced performance compared with previously reported conventional designs.

Keywords Back-surface field · BaSi₂ · CIGS · Photovoltaic · SCAPS-1D · Solar cell

1 Introduction

Solar cells are required for efficient, cost-effective, and highly proficient conversion of sunlight to electrical energy at low cost. Current energy needs worldwide indicate the requirement for cost-effective, efficient power conversion using copper indium gallium diselenide [Cu(In,Ga)Se₂, CIGS]-based thin-film solar cells (TFSCs). The conversion

efficiency of CIGS solar cells is sufficiently high, while their outdoor performance is admirable [1]. Moreover, this chalcogenide material represents a perfect candidate material for the *p*-type absorber layer in CIGS-based solar cells [2]. Although CIGS suffers from high manufacturing cost, it provides better performance than other thin-film solar cell technologies such as CdTe or a-Si:H [3]. This high cost is due to the use of the rare metals gallium (Ga) and indium (In). Due to their scarcity, the CIGS layer should be as thin as possible, to minimize the cost of the absorber material [4].

CIGS comprises four elements, namely copper (Cu), indium (In), gallium (Ga), and selenium (Se). A bandgap of 1.1 eV [5] and an absorption coefficient of 10^5 cm^{-1} [6] have been reported for CIGS. The bandgap (E_g) of Cu(In_{1-x}Ga_x)Se₂ depends on the Ga concentration (*x*) according to Eq. (1) [7],

$$E_g = 1.06 + 0.39238x + 0.24762x^2. \quad (1)$$

✉ Md. Mahabub Alam Moon
mmamoon9357@gmail.com

¹ Department of Electrical and Electronic Engineering, Begum Rokeya University, Rangpur, Rangpur 5400, Bangladesh

² Institute of Electronics, Atomic Energy Research Establishment, Bangladesh Atomic Energy Commission, Dhaka 1207, Bangladesh

³ Solar Energy Laboratory, Department of Electrical and Electronic Engineering, University of Rajshahi, Rajshahi 6205, Bangladesh

Since the value of E_g depends on the proportion of Ga in the CIGS layer, it can be easily tailored from 1.02 to 1.69 eV by varying the Ga/In ratio in the CIGS material [6, 8].

The experimental efficiency of CIGS thin-film solar cell (TFSCs) has been reported to be 19.2% [9], 19.9% [10], and 23.4% [11], while based on numerical analysis, the theoretical efficiency has been reported to be 21.3% [5], 22.02% [12], 22.1% [13], and 22.67% [14]. The optimum value for the thickness of the CIGS absorber layer has been found to be 1 μm [5], 1.4 μm [12], 2 μm [14], and 2.5 μm [13]. With a 1- μm -thick CIGS absorber layer and an additional 1- μm -thick Si absorber layer, an efficiency of 21.3% was confirmed [5]. In addition, an efficiency of 24.45% has also been achieved by using a 1- μm -thick CIGS absorber layer with an additional 0.3- μm SnS BSF layer [15]. An efficiency of 24.55% was obtained by using a CIGS absorber layer with a thickness of 2.2 μm and bandgap of 1.9 eV [2]. Besides, a conversion efficiency of 26% was reported for a 3- μm -thick CIGS absorber layer with a 0.03- μm ZnS buffer layer [16]. When using a 4- μm CIGS absorber layer with E_g of 1.15 eV, an efficiency of 26.15% was found [17]. The efficiency of fabricated CIGS solar cells was increased from 9% to 14% by the use of a molybdenum selenide (MoSe_2) BSF layer [18]. The theoretical efficiency of CIGS solar cells was enhanced from 19.48% to 24% [19] and from 17.99% to 25.29% [20] by using a tin selenide (SnSe_2) and tin sulfide (SnS) BSF layer, respectively. This review of the literature indicates that the CIGS absorber layer is usually 1–4 μm thick [2, 5, 12–17, 20]. It is obvious that the cost of the solar cell will increase as the thickness of the CIGS absorber layer is increased. In this regard, as a potential candidate material for the back-surface field (BSF) layer, the emerging semiconducting barium silicide (BaSi_2) could be utilized in CIGS-based solar cells to reduce the thickness of the CIGS absorber layer.

BaSi_2 comprises barium (Ba) and silicon (Si). Barium has an atomic number of 56 and is found in period 6 of the Periodic Table, having soft elemental form with silvery-gray color [21]. The unit cell of the semiconductor depends on silicon (Si), which has an atomic number of 14 and is found in period 3, having four outer shell electrons. Orthorhombic BaSi_2 has been applied in TFSC applications because of its wide bandgap from 1.1 to 1.35 eV and excellent stability at standard pressure and temperature [22]. Ba and Si are Earth-abundant materials, so the incorporation of a thin film of BaSi_2 could greatly reduce the fabrication cost of CIGS solar cells. The absorption coefficient of BaSi_2 is about 30 [23] to 40 [24] times greater than that of single-crystalline silicon (c-Si). Based on both experimental and theoretical approaches [21–24], BaSi_2 is considered to be a perfect absorber material for solar cells. The potential of BaSi_2 material in TFSCs has been explored [25] based on its fundamental properties.

In the work described herein, BaSi_2 material is exploited for the first time as a BSF layer in a CIGS solar cell. The influence of including such a BSF is explored based on a comparison with similar materials. The results of this work reveal that BaSi_2 is the best material for the BSF layer of CIGS solar cells when compared with other BSF-aided structures [5, 18–20]. In addition, fluorine-doped tin oxide (FTO) is employed for the window layer of the CIGS solar cell.

2 Device and simulation

Figure 1a shows the conventional structure (Al/ZnO/CdS/CIGS/Mo) of the CIGS solar cell with a p -type CIGS absorber having E_g of 1.1 eV [5] and n -type CdS buffer layer having E_g of 2.4 eV [26], creating a junction between them, while the bandgap of the ZnO window layer is 3.3 eV [27]. The proposed TFSC CIGS configuration (Al/SnO₂/CdS/CIGS/ BaSi_2 /Mo) is depicted in Fig. 1b, including a thin, 0.3- μm BaSi_2 BSF layer to significantly reduce the material cost without compromising the performance. The BaSi_2 BSF layer with E_g of 1.3 eV [22] is inserted between the CIGS layer and the molybdenum (Mo) back contact layer. Furthermore, the traditional ZnO window layer is changed to a FTO (SnO₂:F) layer having E_g of 3.6 eV [28]. This new solar cell design comprises a 0.8- μm -thick p -type ($N_A = 1 \times 10^{18} \text{ cm}^{-3}$) CIGS absorber layer, a 0.3- μm -thick p^+ -type ($N_A = 5 \times 10^{18} \text{ cm}^{-3}$) BaSi_2 BSF layer on the back Mo layer, a 0.05- μm -thick n -type CdS buffer layer, and a 0.05- μm -thick SnO₂ window layer. Aluminum (Al) is used as the front grid contact, having a metal surface work function (WF) of 4.06 eV [3]. The molybdenum (Mo) back contact layer has a WF of 4.95 eV [14].

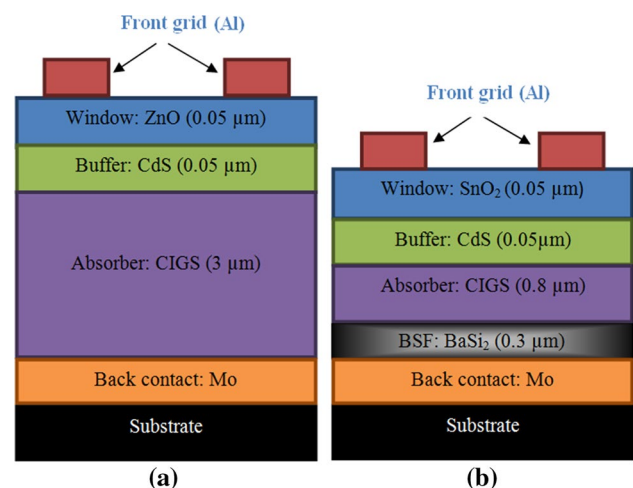


Fig. 1 The CIGS-based solar cell structures: **a** the conventional cell, **b** the proposed cell

For the design and understanding of chalcogenide-based solar cells, numerical simulations have been of great importance through the years. In this work, the solar cell capacitance simulator-one dimensional (SCAPS-1D) [28] simulation program is employed. SCAPS is based on three important equations of semiconductor physics: Poisson's equation, and the continuity equations for free electrons and free holes. In SCAPS, almost all the parameters can be graded, including the bandgap, electron affinity, mobility, doping, thermal speed, effective density of states, etc. Along with the standard test condition (STC) for the illumination spectrum (global air mass, AM 1.5G), a variety of other illumination spectra, including AM0, AM1.5D, monochromatic, white, etc., are available.

Table 1 presents the values for the parameters of the layers used in the solar cells that were input into the simulation.

The amount of current passing through the heterojunction is determined by the band alignment. The energy band diagram for the proposed cell is illustrated in Fig. 2 based on the energy band panel data obtained after the SCAPS simulation, illustrating the bandgap and layer thickness of each material. From Fig. 2, one can easily identify the band bending that occurs between the BaSi_2 and CIGS junction due to the different doping concentration levels used in this study.

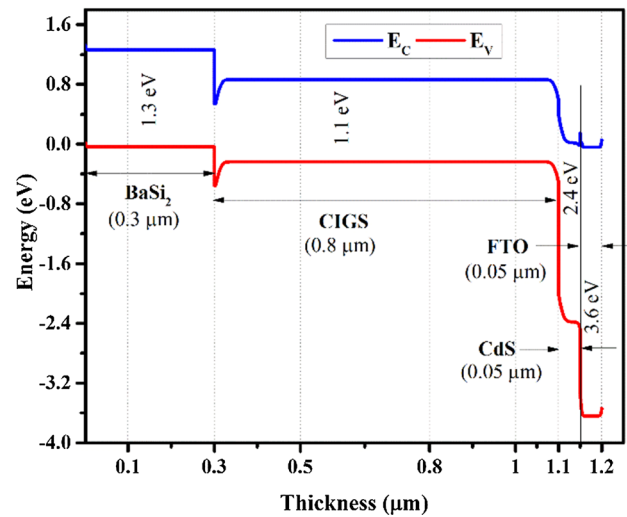


Fig. 2 The energy band diagram of the proposed CIGS solar cell with a BaSi_2 BSF layer

Table 1 The materials properties used in the simulation

Parameter (unit)	n^+ -Type window (ZnO)	n^+ -Type window (SnO_2)	n -Type buffer (CdS)	p -Type absorber (CIGS)	p^+ -Type BSF (BaSi_2)
Thickness (μm)	0.05	0.05	0.05	0.8*	0.3*
Bandgap (eV)	3.3 [2, 5, 28]	3.6 [28]	2.4 [29]	1.1 [5, 7]	1.3 [22]
Electron affinity (eV)	4.6 [5, 28]	4 [28]	4.4 [5, 7]	4.2 [7]	3.3 [22]
Dielectric permittivity (relative)	9 [2, 5, 28]	9 [28]	10 [29]	13.6 [5, 7]	11.17 [30]
CB effective DOS (cm^{-3})	2.2×10^{18} [28]	2.2×10^{18} [28]	2.2×10^{18} [29]	2.2×10^{18} [5, 23]	2.6×10^{19} [22]
VB effective DOS (cm^{-3})	1.8×10^{19} [28]	1.8×10^{19} [28]	1.8×10^{19} [7]	1.8×10^{19} [7]	2×10^{19} [22]
Electron thermal velocity (cm/s)	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7
Hole thermal velocity (cm/s)	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7
Electron mobility ($\text{cm}^2/\text{V s}$)	100 [2, 28]	100 [28]	100 [5, 7]	100 [2, 5]	820 [22]
Hole mobility ($\text{cm}^2/\text{V s}$)	25 [2, 28]	25 [28]	25 [5, 7]	25 [2, 7]	100 [22]
Donor density, N_D (cm^{-3})	1×10^{19}	1×10^{19}	1×10^{18}	0	0
Acceptor density, N_A (cm^{-3})	0	0	0	1×10^{18} *	5×10^{18}
Defect type	SA	SA	SA	SD	SD
Defect density (cm^{-3})	1×10^{15}	1×10^{15}	1×10^{15}	1×10^{14} *	1×10^{15}

CB conduction band, VB valence band, DOS density of states, SA single acceptor, SD single donor

*A variable field

This diagram confirms the possibility that the proposed solar cell will be able to achieve the desired performance.

3 Results and discussion

To simulate the optoelectronic characteristics of the CIGS solar cell, SCAPS software is applied to reveal the effect of decreasing the thickness of the absorber layer (CIGS). The numerical results also illustrate the effect of the introduction of the new BaSi_2 BSF layer on the device performance. The impact of varying the thickness of the CIGS layer, the defects, and the temperature on the photovoltaic parameters was studied for both layouts, under irradiation by the global air mass (AM) 1.5G solar spectrum at temperature of 300 K and incident solar power of 1000 W/m^2 . First, the series and shunt resistances of the solar cell are considered to be zero and infinitely large, respectively, which reflects the case of an ideal solar cell.

3.1 The effect of the thickness of the CIGS absorber layer

3.1.1 Case I: conventional design (without BaSi_2)

The conventional layout of a CIGS solar cell (Fig. 1a) is first examined analytically. The findings are evaluated by comparison with the results for theoretically modeled [5, 31] and experimentally fabricated [9, 10, 32–34] CIGS cells. The impact of the thickness of the p -type CIGS absorber on the photovoltaic (PV) performance of the cell is shown in Fig. 3, where the thickness of the absorber layer is varied from 0.1 to $3 \mu\text{m}$. In this case, the parameter values for the different layers as specified in Table 1 are applied at a temperature of 300 K under 1-sun illumination.

A huge number of photons incident from the solar illumination are absorbed when the thickness of the CIGS absorber layer is increased, thereby increasing the efficiency (η). Indeed, the efficiency is 7.09% and 25.1% for the absorber with thickness of 0.1 and $3 \mu\text{m}$, respectively. For the $3\text{-}\mu\text{m}$ -thick CIGS absorber, the fill factor (FF) is 85.1%, the open-circuit voltage (V_{OC}) is 0.649 V, and the short-circuit current density (J_{SC}) is 40.4 mA/cm^2 , in comparison with analogous reference values of 85% for the FF [16, 35], 0.64–0.69 V for V_{OC} [9, 10, 31, 32], and $\sim 40 \text{ mA/cm}^2$ for J_{SC} [34]. Figure 3 presents the increase in the efficiency with increasing thickness. Also, V_{OC} and J_{SC} varied in proportion to the thickness of the absorber layer. The quantum efficiency (QE) is the ratio of the output charge carrier to the absorbing photons. The variation of the QE of the basic

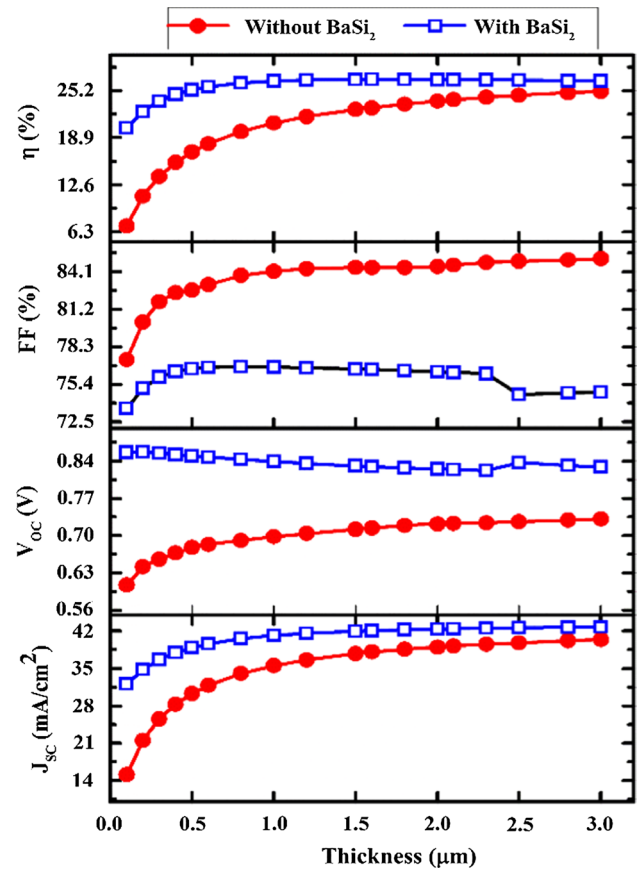


Fig. 3 The effect of varying the CIGS thickness with and without the BaSi_2 BSF layer

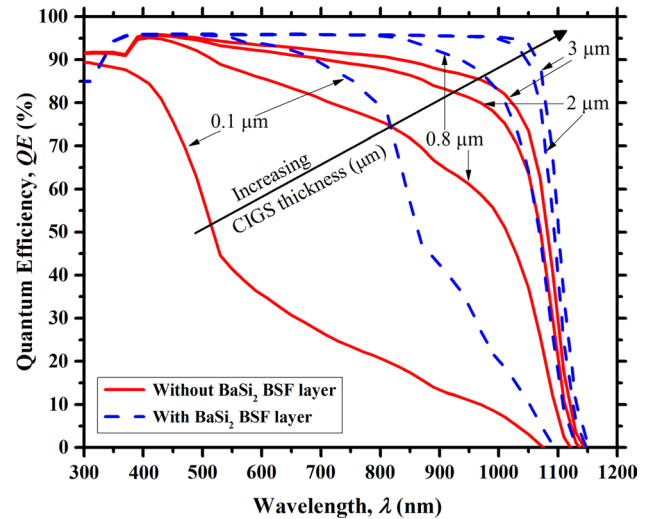


Fig. 4 The variation of Q with wavelength (λ) for the CIGS solar cell

CIGS cell with respect to the thickness of the CIGS layer is shown in Fig. 4.

3.1.2 Case II: the CIGS solar cell with a BaSi₂ BSF layer

A high impurity doping concentration at the back side of the solar cell can prevent the strong recombination of the minority charge carriers (electrons) at the metallic back contact layer. This can be accomplished by introducing a back-surface field (BSF) layer with a doping concentration that is greater than that in the active absorber layer. In the present study, an ultrathin BaSi₂ layer is incorporated to act as such a BSF layer (with doping of $5 \times 10^{18} \text{ cm}^{-3}$), adjacent to the CIGS absorber layer (with doping of $1 \times 10^{18} \text{ cm}^{-3}$). The interface between the p^+ -type BaSi₂ and the p -type CIGS behaves like a p - n junction, forming an electric field that acts as a barrier to the flow of minority carriers to the rear surface. By reflecting the minority carriers, the BaSi₂ BSF layer will thus increase the short-circuit current while decreasing the dark current. The incorporation of the BSF layer thereby minimizes the surface recombination rate and thus enhances the performance of the solar cell.

In this study, the parameter values indicated in Table 1 are applied at a temperature of 300 K under 1-sun illumination. First, the thickness of the BaSi₂ layer is varied from 0.1 to 1 μm . This initial assessment reveals that the efficiency of the cell becomes stable in the range from 0.3 to 1 μm . Hence, a 0.3- μm -thin BaSi₂ BSF layer is considered optimal and applied in the remaining investigation. The traditional CIGS solar cell with a 0.8- μm absorber layer shows an efficiency of 19.72%, but after addition of the 0.3- μm thin p^+ -type BaSi₂ layer, the modified CIGS solar cell exhibits an efficiency of 26.24% for the same thickness of 0.8 μm for the CIGS layer, revealing a great enhancement in efficiency for the same absorber layer thickness. This result is in good agreement with experimental works [18, 36, 37] as well as theoretical research in literature [5, 19, 20, 38–40]. The key reason for this is that both BaSi₂ and CIGS act as absorbers (with a total thickness of 1.1 μm) in the solar cell, whereby the BaSi₂ layer helps to increase the absorption of photon energy by the cell. A large number of photons can thus be captured and more electron–hole pairs generated accordingly. Thus, when moving from the conventional cell design to that proposed herein, J_{SC} increases from 34.1 to 40.559 mA/cm^2 while V_{OC} also increases from 0.691 to 0.8429 V. Very similar behavior of the J_{SC} and V_{OC} parameters was observed in Refs. [2, 5, 6]. Similarly, on incorporation of the BSF layer, J_{SC} increases from 29.25 to 33.47 mA/cm^2 [19] and V_{OC} from 0.62 to 0.79 V [20].

All of these effects enhance the efficiency of the solar cell, greatly improving the PV performance of the traditional CIGS-based cell. However, the fill factor decreases from 83.8% to 76.763%. This occurs because both V_{OC} and J_{SC} increase, whereas V_{MPP} and J_{MPP} do not increase accordingly but rather decrease. Thus, the FF clearly reduces according

to Eq. (2) [41]. However, the FF still retains a good value for solar cell operation.

$$\text{FF} = \frac{V_{\text{MPP}} \times J_{\text{MPP}}}{V_{\text{OC}} \times J_{\text{SC}}}, \quad (2)$$

where V_{MPP} and J_{MPP} denote the voltage and current, respectively, at the maximum power point. The quantum efficiency (QE) can be calculated as the ratio of the current flowing to the external circuit to the number of incident charge carriers. If an electron–hole pair is generated by each photon with a particular wavelength, the external quantum efficiency (EQE) is equal to 1. The impact of the different thicknesses in the new layout on the QE is shown in Fig. 4. The QE of the CIGS/BaSi₂ solar cell first increases with increasing CIGS layer thickness, based on the increased capture of photons by thicker absorber layers. At certain points in the wavelength range, all the curves start to drop towards zero QE, because each material can only absorb photons in a particular wavelength range of the visible light spectrum.

Indium and gallium are high-cost materials in CIGS-based solar cells [4]. Hence, producing absorbers with greater thickness of CIGS is not recommended. Similarly, minimizing the thickness of the CIGS material can decrease the amounts of Ga and In required, which will ultimately lower the manufacturing cost of such devices. BaSi₂ is cheaper and more widely available than Ga and In, and moreover these results indicate higher efficiency when using the new cost-effective solar cell layout compared with the previous design. The results of this investigation could thus guide manufacturing companies towards more profitable production of CIGS solar cells.

3.2 Impact of the CIGS acceptor density on the performance of the solar cell

The acceptor density (N_{A}) in the CIGS layer is varied from 1×10^{12} to $1 \times 10^{20} \text{ cm}^{-3}$ to investigate its effect on the performance of the CIGS solar cell. In this case, the acceptor density for the BaSi₂ layer is $5 \times 10^{18} \text{ cm}^{-3}$, while the donor density in the CdS layer and SnO₂ layer is kept constant at $1 \times 10^{18} \text{ cm}^{-3}$ and $1 \times 10^{19} \text{ cm}^{-3}$, respectively. Figure 5 shows the effect of varying the acceptor density in the CIGS layer. The results reveal that changing the N_{A} value in the CIGS material from 1×10^{12} to $1 \times 10^{20} \text{ cm}^{-3}$ increases V_{OC} from 0.704 to 0.963 V, J_{SC} from 40.18 to 40.5 mA/cm^2 , the FF from 39.79% to 83.9%, and the efficiency from 16.7% to 32.7%. Varying the acceptor density in the CIGS absorber layer changes all the PV parameters in an analogous way to that reported in Refs. [5, 16]. A very similar phenomena has been reported when changing the N_{A} value in the CdTe absorber layer with using a vanadium pentoxide (V₂O₅) BSF in CdTe solar cells [38].

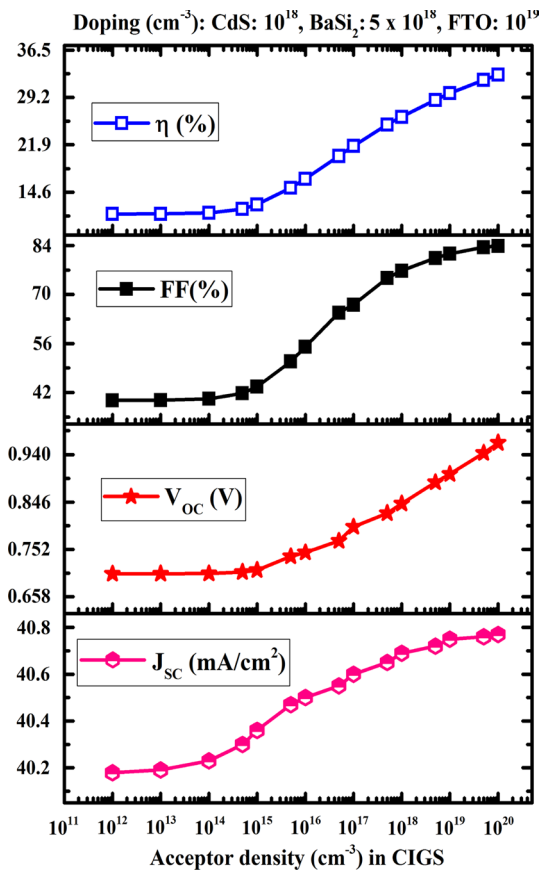


Fig. 5 The impact of the acceptor density in the CIGS layer on the characteristics of the solar cell

Considering the cost and growth rate of device fabrication, the optimal value of the acceptor impurity density in the CIGS material for the proposed design is found to be $1 \times 10^{18} \text{ cm}^{-3}$.

3.3 The impact of the bulk defect density in the CIGS absorber layer on the proposed solar cell

To study the effect of the bulk defect density in the CIGS layer on the photovoltaic performance of the solar cell, the single-donor-type bulk defect density is varied from 1×10^{12} to $5 \times 10^{17} \text{ cm}^{-3}$, while keeping the defect densities of all the other layers constant at the values specified in Table 1. Figure 6 shows the impact of the different defect densities on the performance parameters of the solar cell. The values of V_{OC} and J_{SC} decrease as the defect density is increased; J_{SC} drops from 40.87 to 28.5 mA/cm^2 , while V_{OC} falls from 0.97 to 0.516 V. The conversion efficiency thus falls from 34.27% (for a defect density of $1 \times 10^{12} \text{ cm}^{-3}$) to 9.11% (for a defect density of $5 \times 10^{17} \text{ cm}^{-3}$). As the defect density is increased, the probability of carrier recombination before the carriers

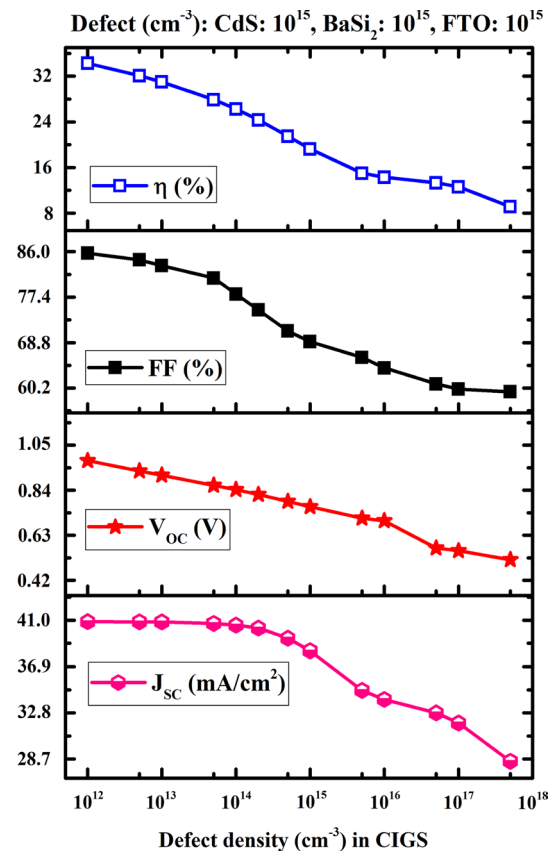


Fig. 6 The effect of varying the defect density in the CIGS layer on the characteristics of the proposed solar cell

can reach the junction and generate current is increased. In other words, the higher the defect density, the lower the rate of creation of electron–hole pairs. As described in Refs. [5, 6, 13, 15], defects can easily deteriorate the photovoltaic performance, thereby diminishing the efficiency of the solar cell. In this regard, to improve the output of the proposed cell, the defect density must be limited to as low as possible and should not exceed the limit of $1 \times 10^{14} \text{ cm}^{-3}$.

3.4 The effect of the resistances on the PV performance of the proposed solar cell

The effect of the series (R_s) and shunt resistances (R_{sh}) on the photovoltaic device is studied using SCAPS-1D software. The R_s of the solar cell results from the bulk resistance, the front and back metallic contact resistance, and the circuit terminal resistance [5]. The key effect of the series resistance is to diminish the fill factor [42], meaning that a FF of 100% cannot be achieved. Lower series and higher shunt resistances must be achieved to increase the efficiency. The values of these resistances also affect V_{OC} and J_{SC} . Using SCAPS-1D, the value of R_s is varied to observe the impact on the performance of both the traditional and proposed solar cells.

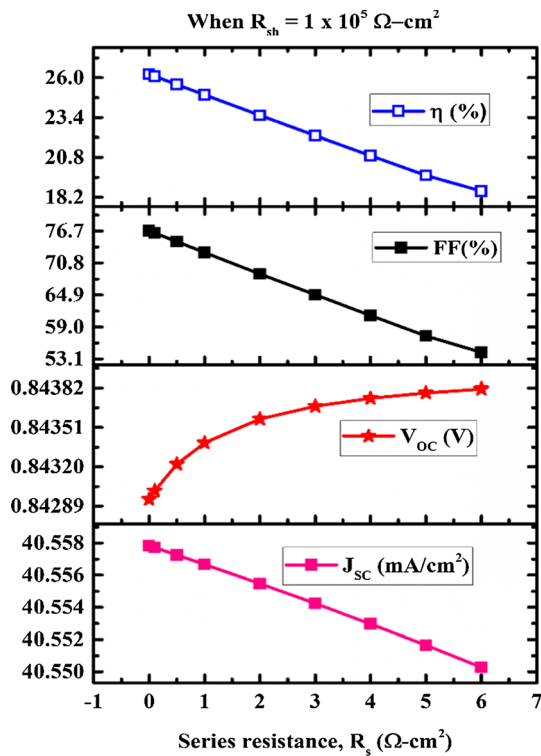


Fig. 7 The impact of varying R_s on the characteristics of the proposed cell

The value of R_s is varied from 0 (ideal case) to $6 \Omega \text{ cm}^2$, while the shunt resistance (R_{sh}) is fixed at $10^5 \Omega \text{ cm}^2$. Figure 7 illustrates the effect of changing R_s on the proposed cell including the BaSi_2 BSF layer, indicating that the efficiency of the solar cell is significantly degraded by increasing R_s , as also observed in literature [5, 16]. These results reveal a decrease in the efficiency from 26.24% to 18.59% when changing R_s over the mentioned range.

Fixing the series resistance, R_s at $0.5 \Omega \text{ cm}^2$, the value of R_{sh} is now varied from 10 to $1 \times 10^7 \Omega \text{ cm}^2$. Figure 8 illustrates the results when changing the R_{sh} value of the cell as mentioned, showing that the efficiency of the proposed solar cell increases with increasing R_{sh} . The efficiency is found to increase from 25.08% to 25.56% on increasing the R_{sh} value of the proposed solar cell including the BaSi_2 BSF layer.

3.5 The impact of the cell operating temperature

The performance of the CIGS solar cells with and without the BaSi_2 BSF layer is examined while varying the cell temperature (T) from 273 to 473 K, with all other parameters kept constant as specified in Table 1. Figure 9 illustrates the effect of such variation of the temperature on J_{sc} , V_{oc} , the FF, and the efficiency for both cases, viz. with and without the BaSi_2 BSF layer. The characteristics of the solar cell depend strongly on temperature, since V_{oc} , J_{sc} , η

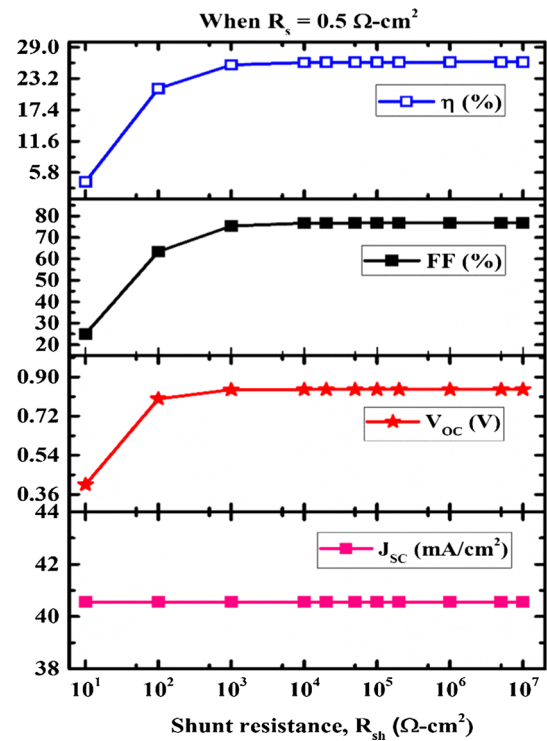


Fig. 8 The impact of varying R_{sh} on the characteristics of the proposed cell

(efficiency), and the FF are temperature-dependent parameters. Note that V_{oc} varies inversely proportionally with the temperature, because the saturation current decreases rapidly at increasing temperature, due to the increase in the reverse saturation current. V_{oc} is a saturation current-dependent parameter. The short-circuit current density (J_{sc}) remains almost constant with increasing temperature. The proposed solar cell layout also shows a good response to the temperature variation. When the temperature is low, the efficiency is high enough. At a temperature of 273 K, the efficiency is 25.81%, but it reduces to 17.9% at 473 K. At much higher temperature, the efficiency was also degraded in Refs. [2, 5, 21].

The lower efficiency observed at higher temperatures is due to the temperature dependence of parameters such as the carrier concentrations, electron and hole mobilities, and energy bandgaps of the materials [32]. The E_g values become unbalanced as T is increased, and the resulting electron–hole recombination in the unstable bandgaps diminishes the conversion efficiency [14]. For both cases, viz. with and without the BSF, the short-circuit current density remains almost constant. With the BaSi_2 BSF layer, J_{sc} is always $\sim 6.6 \text{ mA/cm}^2$ higher than in the conventional structure across the entire temperature variation. It is obvious from Fig. 9 that both the cell efficiency and open-circuit voltage decrease linearly for the case of the basic CIGS cell.

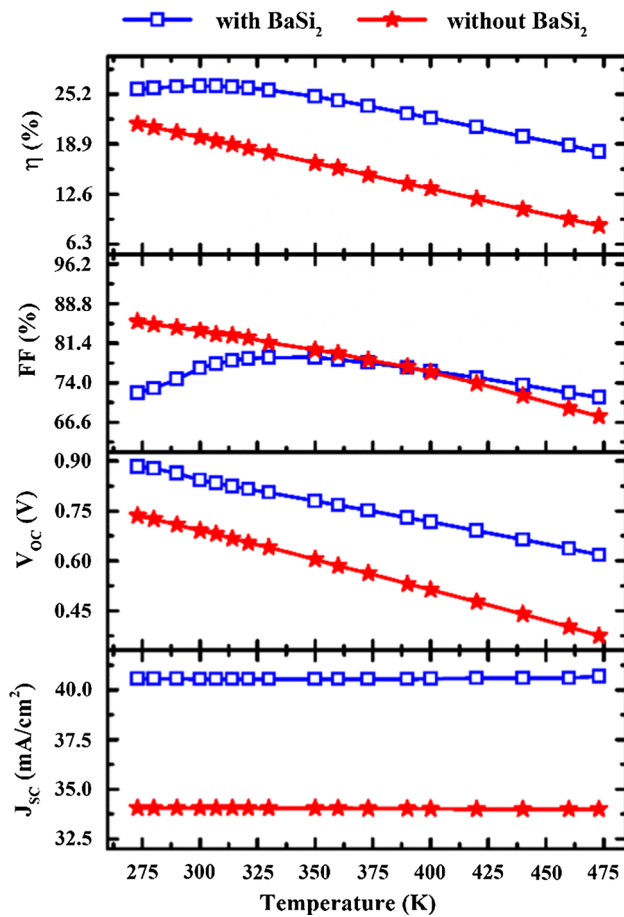


Fig. 9 The temperature dependence of the PV parameters of the cell

However, on inclusion of the BaSi_2 BSF layer, these two parameters reduce in a less linear fashion, clearly confirming the overall improvement achieved by the addition of the ultrathin BaSi_2 BSF layer to the conventional CIGS solar cell.

3.6 The J – V characteristic of the CIGS-based solar cells

Figure 10 displays the current–voltage (J – V) characteristic curves of the CIGS solar cells. Table 2 summarizes the photovoltaic parameters of the proposed solar cell structure with a 0.8- μm -thick CIGS absorber layer, which is seen to be superior to the performance of the conventional CIGS solar cell with either a 0.8- μm - or 3- μm -thick absorber, as observed in the table. However, the FF is reduced by about 8% in the proposed cell (with a 0.8- μm -thick CIGS absorber) compared with the conventional cell (with a 3- μm -thick CIGS absorber). The reason behind this decrement is explained in Sect. 3.1 (case II), although the value is still a good result.

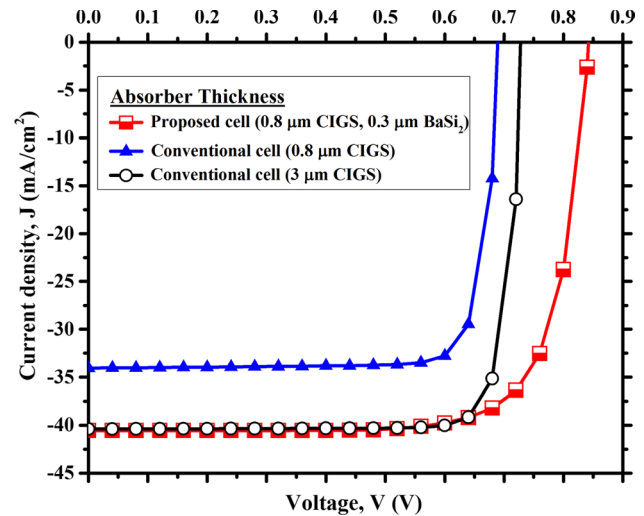


Fig. 10 The J – V characteristic of the CIGS-based solar cells

Table 3 presents a comparison of the PV parameters obtained by some theoretical and experimental analyses of CIGS solar cells, in comparison with the results of the current work. It is evident from Table 3 that, with only a thin, 800-nm CIGS absorber layer, the TFSC cell proposed herein could be commercially more attractive compared with the other CIGS-based structures mentioned in Table 3. Also, the structure provides a route to decrease the absorber material cost of CIGS-based solar cells. The proposed solar cell is thus superior to the other CIGS solar cell structures [2, 5, 9–20].

3.7 An evaluation of the potential of the BaSi_2 BSF in CIGS solar cells

Materials such as SnS , SnSe_2 , Si , MoSe_2 , etc. have been employed as a BSF layer in CIGS solar cells and analyzed either theoretically or experimentally. The beneficial effect of using a BaSi_2 BSF in CIGS solar cells and a comparative summary versus similar materials are presented in Table 4. Among the mentioned BSF layers, BaSi_2 exhibits a relatively higher absorption coefficient. Considering its ready availability in the Earth's crust, BaSi_2 enables lower-cost fabrication compared with SnS , SnSe_2 , or MoSe_2 . Moreover, its bandgap can be adjusted to the nearly optimal value of 1.4 eV by introducing strontium (Sr) impurities [21]. It also possesses a long minority-carrier diffusion length and lifetime [21, 25]. The simulation results clearly verify the potential of such a BaSi_2 BSF layer to enhance the PV performance of CIGS solar cells. Furthermore, based on the results presented throughout the current paper and considering the advantages summarized in Table 4, BaSi_2 can be considered to be the best material for the BSF layer of CIGS-based TFSCs.

Table 2 A summary of the simulation results

Cell type	CIGS thick- ness (μm)	BaSi ₂ thick- ness (μm)	J_{SC} (mA/cm ²)	J_{MPP} (mA/cm ²)	V_{OC} (V)	V_{MPP} (V)	FF (%)	η (%)
Conventional	3.0	–	40.40	38.700	0.731	0.649	85.1	25.10
Conventional	0.8	–	34.10	32.296	0.691	0.610	83.8	19.71
Proposed	0.8	0.3	40.56	37.060	0.843	0.708	76.8	26.24

Table 3 The photovoltaic performance of the proposed cell in comparison with other reported CIGS solar cells

No.	Type of research	CIGS layer thickness (nm)	V_{OC} (V)	J_{SC} (mA/cm ²)	FF (%)	η (%)	Ref.
1	Experimental	2000	0.671	34.90	77.60	18.10	[32]
2	Experimental	1000	0.689	35.71	78.12	19.20	[9]
3	Experimental	2200	0.690	35.50	81.20	19.90	[10]
4	Experimental	–	0.741	37.80	80.60	22.60	[33]
5	Experimental	–	0.734	39.58	80.40	23.85	[34]
6	Theoretical	3000	0.640	34.60	79.50	17.70	[31]
7	Theoretical	1000	0.743	34.47	83.09	21.30	[5]
8	Theoretical	1000	0.780	38.66	80.00	24.45	[15]
9	Theoretical	2200	1.020	27.66	86.88	24.55	[2]
10	Theoretical	3000	0.824	36.90	85.50	26.00	[16]
11	Theoretical	4000	0.800	38.20	75.00	26.15	[17]
12	Theoretical	2500	0.798	38.66	85.15	26.30	[35]
13	Theoretical	800	0.843	40.56	76.80	26.24	[^a]

[^a] The cell proposed herein**Table 4** The influence of the back-surface field (BSF) layer in comparison with similar research

No.	Type of research	Absorber	BSF	η without BSF (%)	η with BSF (%)	Enhancement in η (%)	Ref.
1	Experimental	Si	ZnS	6.40	11.02	72.19	[36]
2	Experimental	Si	Al	12.96	13.75	6.09	[37]
3	Experimental	CIGS	MoSe ₂	9	14	55.55	[18]
4	Theoretical	CdTe	V ₂ O ₅	19.58	23.50	20.00	[38]
5	Theoretical	CZTS	CZTS	12.05	14.11	17.09	[39]
6	Theoretical	ZnTe	Sb ₂ Te ₃	7.14	18.33	156.72	[40]
7	Theoretical	CZTSSe	SnS	12.30	17.25	40.24	[43]
8	Theoretical	CIGS	Si	16.39	21.30	29.96	[5]
9	Theoretical	CIGS	SnSe ₂	19.48	24	23.20	[19]
10	Theoretical	CIGS	SnS	17.99	25.29	40.57	[20]
11	Theoretical	CIGS	BaSi ₂	19.71	26.24	33.13	[^a]

[^a] The cell proposed herein

4 Conclusions

SCAPS-1D simulation is applied to study the behavior of thin-film solar cells based on CIGS absorber material. The effects of the thickness, doping concentration, bulk defect density, resistances, and temperature on the optoelectronic output parameters of the cells are thoroughly

investigated. A thin, 0.3- μm BaSi₂ layer is inserted into the basic CIGS solar cell structure of Al/ZnO/CdS/CIGS/Mo with a ZnO window, resulting in an alternative structure of Al/FTO/CdS/CIGS/BaSi₂/Mo with a FTO window. The photovoltaic parameters of the modeled cells are evaluated numerically and compared with those reported in other theoretical and experimental studies. The addition of the back-surface field (BSF) layer made from BaSi₂ results

in a significant enhancement of the current, voltage, and efficiency while also notably reducing the thickness of the absorber material. BaSi_2 can thus be said to represent a favorable material for the BSF in CIGS-based solar cells. The optimized thickness of the FTO window, CdS buffer, CIGS absorber, and BaSi_2 BSF layers is found to be 0.05 μm , 0.05 μm , 0.8 μm , and 0.3 μm , respectively. The acceptor impurity concentration in the CIGS layer is chosen to be $1 \times 10^{18} \text{ cm}^{-3}$. For a certain number of defects, this structure presents an efficiency (η) of 26.24% along with a V_{OC} of 0.843 V, J_{SC} of 40.56 mA/cm^2 , and FF of 76.76%. The results of this work provide design guidelines for the introduction of an ultrathin BaSi_2 BSF layer into a conventional CIGS solar cell to enhance its overall performance while significantly reducing the cost of the absorber material.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflicts of interest.

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